Department of Physics
Indian Institute of Technology Madras

B.Tech. Laboratory Manual
PH1030 (Jan-May 2026)

List of experiments:

1. Young's modulus of wood using a strain gauge
2. Radiation from a hot filament – Stefan's Law
3. Variable 'g' pendulum
4. Transmission grating
5. Mapping equipotential lines
6. Ultrasonic Interferometer – Velocity of ultrasonic waves
7. Wavelength of light by interference - Newton's rings
8. Transistor Characteristics
9. Magnetic field along the axis of the coil.

PH1030: Instructions to students (Jan-May 2026)

1. Details about the course:

- Number of experiments: 9
- List of experiments: Given in the laboratory manual
- Batch wise allotment of experiments: Will be announced in the lab

2. Preparation for the experiments:

- Students should procure the laboratory manual from Moodle (https://courses.iitm.ac.in) and 9 sets of stapled laboratory record sheets during the first week when you come to the lab. The 9 sets of record sheets will be used for 9 different experiments and students should use only these sheets. Students should come prepared with the aim, apparatus required for the experiment, circuit/ray/schematic diagram, formula, tabular columns, etc. written down (before coming to the laboratory sessions) on the record sheets-set.

- **Marks will be reduced for those who come to laboratory without preparation**

3. How to carry out the experiments:

- Experiments should be completed on the same day, and marks can be obtained from the teacher at the end of the laboratory session.
- The record sheets set used by the student for the experiment should not be carried out of the laboratory. Failure to return the set will be considered as the experiment being incomplete and no marks will be given.
- Student should inform the concerned teacher and also the laboratory staff (**return the issued experimental accessories to the staff**) before leaving the laboratory.
- **It is mandatory for all students to wear shoes for the laboratory classes.**

4. Repeat classes:

- No repetition of an experiment is allowed except in the event of the student being absent due to medical emergency and unforeseen circumstances. Repeat classes (for various batches) will be held at the end of the experiments-cycle. Please note that we have only limited days for the repeat classes for entire sessions.

5. Evaluation Procedure:

- Each experiment will be evaluated for 30 marks for in-lab activities and 10 marks for "outside the lab hours activities" and the grand total (10x40) will be normalized to 75 marks. The end semester practical examination will be conducted at the end of the cycle and it carries 25 marks.

6. **Error Analysis:** A manual on 'Introduction to Error Analysis' is available on Moodle. Please read through the manual and cited references to correctly perform the error analysis for your measurements.

Graphs

A graph is simply a diagram illustrating the relationship between two quantities, one of which varies as the other is changed. The quantity that is changed is called "independent variable", the other is called the "dependent variable". The following general points should be noted:

1. The scale must not be too small as this leads to loss of precision. The scale should not be too large as it leads to exaggeration of accidental errors.

2. The independent variable is plotted on the horizontal axis and dependent variable is plotted on the vertical axis.

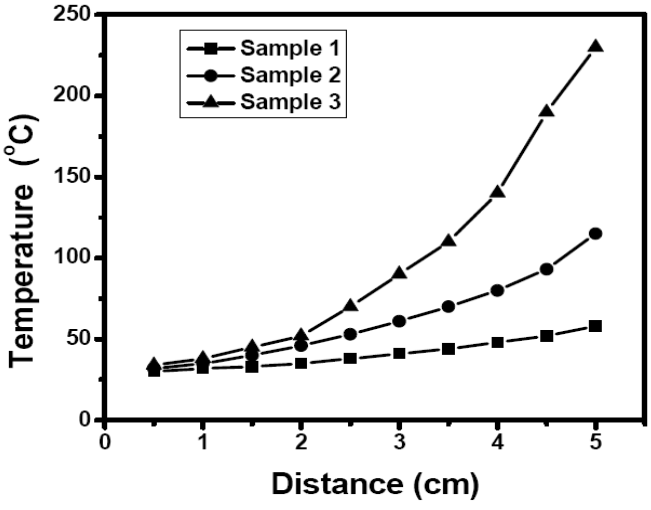
3. The origin need not represent the zero values of variables, but the value at the origin must be clearly marked.

4. The graph should be titled. The axes must have legible captions containing: *a*) standard name of the variable, *b*) its symbol, if such a thing exists, and *c*) standard abbreviation for the unit of measure being used.

5. Numerals representing scale values should be placed outside the axis. Values less than unity should be written as 0.47, not .47. Use of too many zeroes should be avoided. Thus, if the scale numbers are 10,000, 20,000, 30,000 etc., they should be written as 1.0, 2.0, 3.0 with the symbol and unit, say, pressure, represented as P (10^4 N/m^2). Similarly scale numbers such as 0.0001, 0.0002, 0.000, etc., should be written as 1.0, 2.0, 3.0 with the symbol and units, say, pressure, represented as P (10^{-4} N/m^2).

6. An example of a clearly labelled graph is shown below.

Example:



1. Young's modulus of wood using a strain gauge

Aim: To determine the Young's modulus of a half-meter wooden scale

Apparatus: A half meter scale with two identical strain gauges attached to each end of the scale with one strain gauge at the top and the other at the bottom, thick wire, a clamp, a circuit board with appropriate terminals to constitute a Wheatstone network, hanger and slotted weights, digital millivoltmeter, calipers, screw gauge, rheostats, a constant current source and connecting wires.

Young's modulus:

When an external force F is applied along a long bar of length L in a direction perpendicular to the cross-sectional area A, internal forces in the bar resist distortion. As a result, the bar attains an equilibrium when the external force is exactly balanced by the internal forces with a change in length ΔL. The tensile stress is force per unit area (in N/m^2) and the longitudinal strain is the change in length relative to the original length and it is a dimensionless quantity. The ratio of the tensile stress (F/A) to the tensile strain ($\Delta L/L$) is given by

$$Y = \frac{F/A}{\Delta L/L}$$

where Y is the Young's modulus of the bar.

Strain Gauge:

A strain gauge is a transducer whose electrical resistance varies in proportion to the amount of strain in the device. The most widely used gauge is a metallic strain gauge which consists of a very fine wire or, more commonly, metallic foil arranged in a grid pattern. The grid pattern maximizes the amount of metallic wire or foil subject to strain in the parallel direction (Figure 1). The cross-sectional area of the grid is minimized to reduce the effect of other types of strain, such as shear strain and Poisson strain. The grid is bonded to a thin backing, called the carrier, which is attached directly to the test specimen. Therefore, the strain experienced by the test specimen is transferred directly to the strain gauge, which responds with a linear change in electrical resistance.

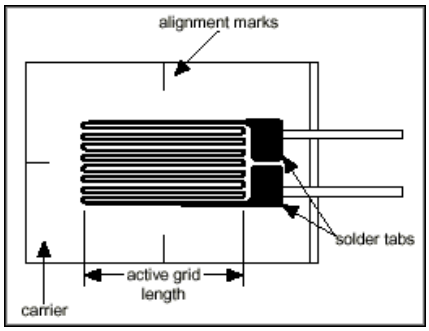


Figure 1: A strain gauge

A fundamental parameter of the strain gauge is its sensitivity to strain, expressed quantitatively as the gauge factor (λ). Gauge factor is defined as the ratio of fractional change in electrical resistance to the fractional change in length (strain). The gauge factor for metallic strain gauge is typically around 2.

Wheatstone Bridge:

Measuring the strain with a strain gauge requires accurate measurement of very small changes in resistance R and such small changes can be measured with a Wheatstone bridge. A general Wheatstone bridge consists of four resistive arms with an excitation voltage, V_{ex} , that is applied across the bridge (Figure 2).

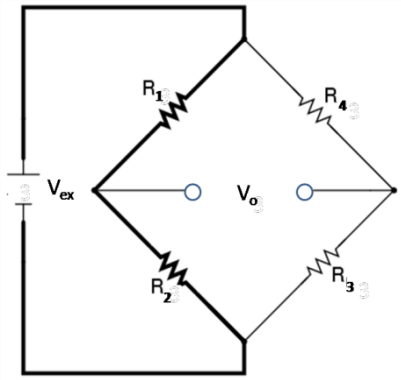


Figure 2: Wheatstone bridge

The output voltage of the bridge, V_o, will be equal to:

$$V_o = \left[\frac{R_3}{R_3+R_4} - \frac{R_2}{R_1+R_2}\right] V_{ex}.$$

From this equation, it is apparent that when $R_1/R_2 = R_4/R_3$, the voltage output V_o will be zero. Under these conditions, the bridge is said to be balanced. Any change in resistance in any arm of the bridge will result in a nonzero output voltage. Therefore, if we replace R_4 in Figure 2 with an active strain gauge, any changes in the strain gauge resistance will unbalance the bridge and produce a nonzero output voltage. If the nominal resistance of the strain gauge is designated as R_G, then the strain-induced change in resistance, ΔR, can be expressed as $\Delta R = R_G \times \lambda \times \text{Strain}$. Assuming that $R_1 = R_2$ and $R_3 = R_G$, the bridge equation above can be rewritten to express V_o/V_{ex} as a function of strain.

Ideally, we would like the resistance of the strain gauge to change only in response to applied strain. However, strain gauge material, as well as the specimen material on which the gauge is mounted, will also respond to changes in temperature. Strain gauge manufacturers attempt to minimize sensitivity to temperature by processing the gauge material to compensate for the thermal expansion of the specimen material for which the gauge is intended. While compensated gauges reduce the thermal sensitivity, they do not totally remove it. By using two strain gauges in the bridge, the effect of temperature can be further minimized. For example, in a strain gauge configuration, one gauge is active, i.e. its resistance changes with strain as

$R_G + \Delta R$, and a second gauge is placed transverse to the applied strain. Therefore, the strain has little effect on the second gauge, called the dummy gauge. However, any changes in temperature will affect both gauges in the same way. Because the temperature changes are identical in the two gauges, the ratio of their resistance does not change, and hence the voltage V_o does not change, and thus the effects of the temperature change are minimized.

The sensitivity of the bridge to strain can be doubled by making both gauges active in a half-bridge configuration. Figure 3 illustrates a bending beam application with one bridge mounted in tension $(R_G + \Delta R)$ and the other mounted in compression $(R_G - \Delta R)$. This half-bridge configuration, whose circuit diagram is also illustrated in Fig.3, yields an output voltage that is linear and approximately doubles the output of the quarter-bridge circuit.

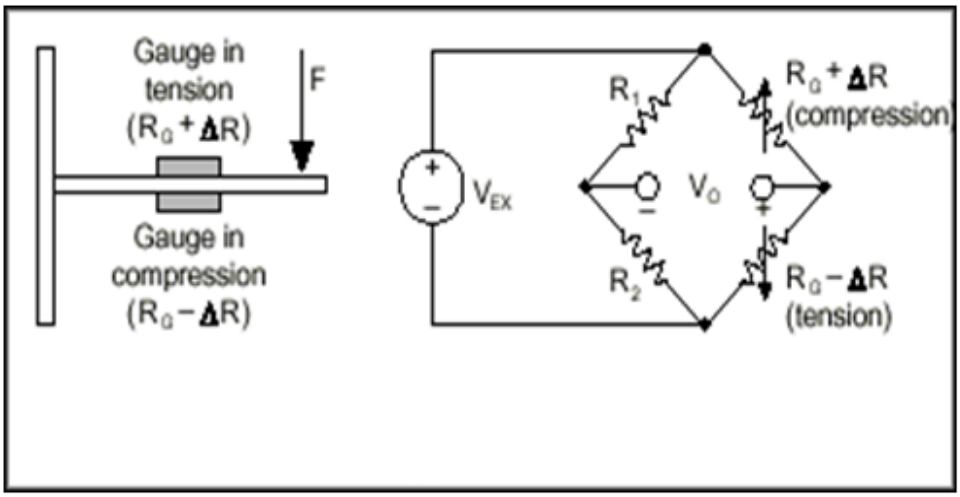


Figure 3.

In this experiment, we aim to determine the Young's modulus of a half-meter wooden bar by loading it with a mass m. For a beam of rectangular cross-section with breadth b and thickness d, and area moment of inertia I, the restoring moment equals $Y \times I/R$, where R is the radius of curvature of the bending beam and Y is the Young's modulus. The Young's modulus (Y) can be calculated by assuming that at equilibrium, the bending moment is equal to the restoring moment. Note that in the experiment, the load (stress) is applied perpendicular to the bar. This leads to a bending of the bar, but if we imagine the bar as a bundle of fibres, then bending is nothing but a set of axial strains that vary along the thickness of the bar: The fibres on the top surface of the bar are under tension, but those on the bottom surface are under compression. Hence, the bending characteristics under an applied transverse stress are related to the Young's modulus and this configuration can be used to determine the latter quantity.

Procedure:

1. Clamp the beam to the table in such a way that the strain gauges are close to the clamped end. Load and unload the free end of the beam a number of times.

2. Make the connections as given in the circuit diagram (Figure 4.)

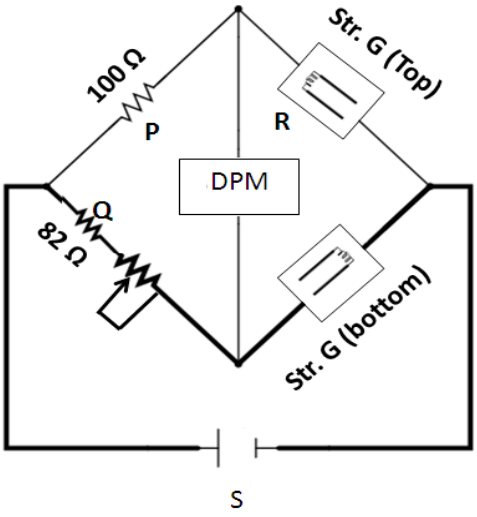


Figure 4

3. Switch on the constant current source and the DPM (Digital Panel Meter).
4. Balance the bridge using the variable resistance box. At this stage the DPM will read 0 or very nearly zero. Note that this is done at no load.
5. Load the beam with a hanger of mass m, suspending it as close to the free end as possible. Note the DPM reading. Note that as you are about to take a reading, the last digit will be changing about the actual steady value. Take a few readings and get the average.
6. Increase the load in steps of m up to $5m$ and take the readings each time.
7. Unload the beam from $5m$ down to zero in steps of m at a time and note the DPM reading each time.
8. To check reproducibility, repeat all the above processes taking readings while loading and unloading in steps of m.
9. Compute the quantity (dV/M), where dV is the unbalanced voltage and M is the load, for each load used. Combine your estimates into a final estimate of dV/M and estimate the uncertainty in this quantity.
10. Draw a graph between M along x-axis and unbalanced voltage dV along y-axis. Plot a straight line with your estimated value of dV/M for visual comparison.
11. Note the distance between the centre of the strain gauges and the point of application of the load (L).
12. Measure the breadth of the beam using slide calipers (b).
13. Measure the thickness of the beam using a screw gauge (d).

14. Young's Modulus of the material of the beam, which is nothing but the stress to strain ratio, is given by the following expression:

$$Y = \left(\frac{6gL\lambda RI}{bd^2\left[1+\left(\frac{R}{P}\right)\right](dV/M)} \right).$$

Here, g is acceleration due to gravity, λ is the gauge factor (to be obtained from the teacher), I is the output current from the source, R is the resistance of strain gauge and P is the fixed resistance shown in Figure 4.

Table 1: Output voltage of Wheatstone bridge under varying load

Load/gm →	0	m	2m	3m	4m	5m
DPM reading						
1) Loading V_1/mV						
2) Unloading V_2/mV						
Mean of V_1+V_2						

Table 2: To find the thickness of the given beam

Least count of the given screw gauge:(LC) _____ *mm*

Sl.No.	Pitch scale Reading (PSR)	Head scale coincidence (HSC)	Head scale reading (HSC x LC)	Total reading (mm) (PSR+HSR)

Table 3: To find the breadth of the given beam using slide calipers

Least count of the given slide calipers:(LC) _____ *cm*

Sl.No.	Main Scale Reading (PSR)	Vernier Scale Coincidence (VSC)	Vernier Scale Reading (VSC x LC) (VSR)	Total reading (cm) (VSR+HSR)

2. Radiation from a hot filament – Stefan's Law

Aim: Verification of Stefan's law

Apparatus: Variable power supply, Low voltage bulb, Digital Voltmeters, Digital Multimeter, Thermopile, Measuring Scale and connecting wires.

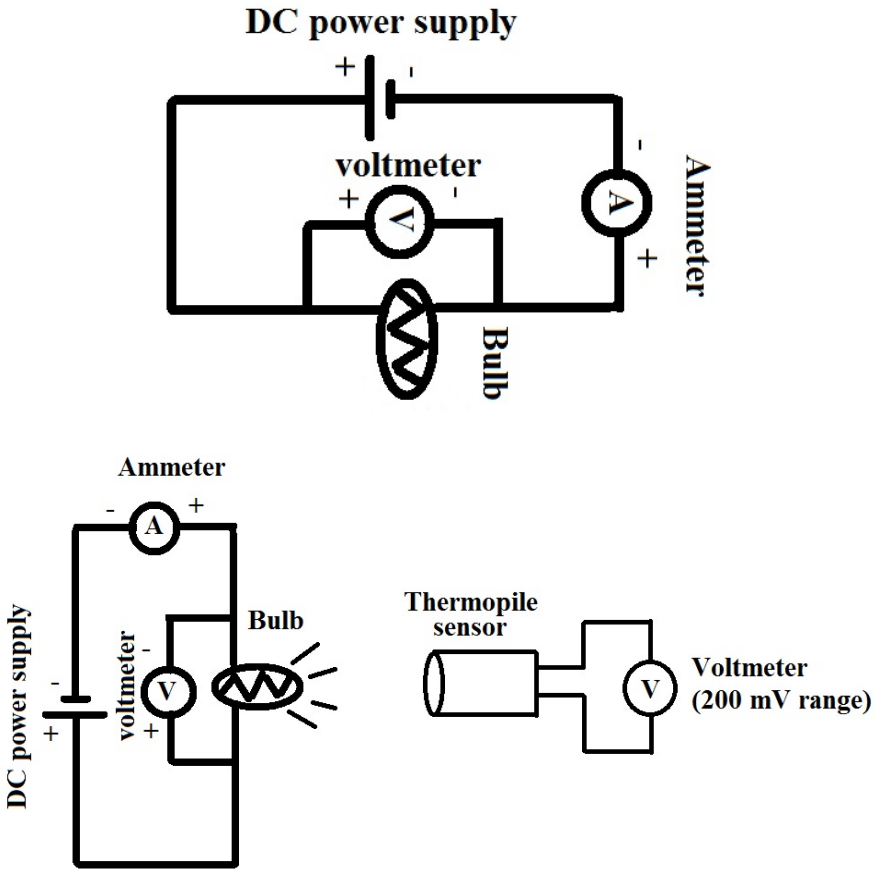


Figure 1: Circuit diagrams for measuring the resistance of the filament (when the bulb is not glowing) and determining filament temperature (when the filament starts glowing).

Theory:

Let the energy emitted by a black body per unit area and unit time at temperature T in the wavelength range λ to $\lambda + d\lambda$ be denoted by $\left(\frac{dL(T,\lambda)}{d\lambda}\right) d\lambda$. This quantity can be integrated over wavelength to yield the total flux density $L(T)$:

$$L(T) = \frac{2\pi^5}{15}\frac{k^5}{c^2h^3}T^4.$$

Here, c = speed of light (3.00 x 10^8 m/s), h = Planck's constant (6.62 x 10^{-34} J·s) and k = Boltzmann's constant (1.381 x 10^{-23} J·K^{-1}). This is called Stefan-Boltzmann law and can be written as

$$L(T, \lambda) = \sigma T^4$$

with σ = 5.67 x 10^{-8} W·m^{-2}·K^{-4}.

The proportionality $L \propto T^4$ is also valid for a 'grey' body, the surface of which shows a wavelength-independent absorption-coefficient of less than one.

To prove the validity of Stefan-Boltzmann's law, we measure the radiation emitted by the filament of an incandescent lamp which represents a 'grey' body fairly well. For a fixed distance between filament and thermopile, the energy flux Φ which hits the thermopile is proportional to $L(T)$:

$$\Phi \propto L(T).$$

Because of the proportionality between Φ and the thermoelectric e.m.f. V_{TC} of the thermopile, we can also write

$$V_{TC} \propto T^4,$$

if the thermopile is at a temperature of zero Kelvin. Since the thermopile is at room temperature T_R, it also radiates due to the T^4 law so that we can write

$$V_{TC} \propto (T^4 - T_R^4).$$

Under the present circumstances, we can neglect T_R^4 compared to T^4, so that we get a straight line with a slope "4" when representing the function $V_{TC} = f(T)$ on a log-log plot:

$$\log(V_{TC}) = 4 \log(T) + \text{constant}.$$

The absolute temperature $T = t + 273$ of the filament is calculated from the measured resistances $R(t)$ of the tungsten filament (t = temperature in centigrade). For the tungsten filament resistance, we have the following temperature dependence:

$$R(t) = R_o(1 + \alpha\, t + \beta\, t^2)$$

where, R_0 = resistance at 0° C, $R(t)$ = resistance at temperature t, α = 4.82 x 10^{-3} K^{-1} and β = 6.67 x 10^{-7} K^{-2}.

The resistance R_0 at 0°C can be found by using the relation:

$$R_0 = \left(\frac{R(t_R)}{1+\alpha \cdot t_R + \beta \cdot t_R^2}\right). \qquad (5)$$

Solving for t in terms of $R(t)$ and using the relation $T = t + 273$ gives:

$$T = 273 + \frac{1}{2\beta}\left[\sqrt{\alpha^2 + 4\beta\left(\frac{R(t)}{R_0} - 1\right)} - \alpha\right] = 273 + \frac{\alpha}{2\beta}\left[\sqrt{1 + \frac{4\beta}{\alpha^2}\left(\frac{R(t)}{R_0} - 1\right)} - 1\right]. \quad (6)$$

$R(t)$ and R_0 are found by applying Ohm's law, e.g. by voltage and current measurements across the filament.

Procedure:

Part 1: Measurement of the resistance of the filament at room temperature.

1. Connect the circuit as shown in Figure 1.
2. Voltage and current should be measured using digital multimeters.
3. Switch ON the DC source. Adjust the voltage (V) to $0.1\,V$ and measure the corresponding current (I). Records these values in the tabular column.
4. Calculate the resistance of the filament (V/I) and use it as $R(t_{\mathrm{room}})$ where t_{room} is the room temperature. Thus determine R_0, the resistance at $0°\mathrm{C}$. Note: The current can be measured for few different voltages (measured in steps of $0.1\,V$) and an average R_0 can be calculated. This measurement should be done at room temperature (i.e. under cold condition) of the filament. Larger current can heat up the coil and therefore modify the actual R_0 at room temperature.

Part 2. Measurement of the resistance at different currents

5. Use the same circuit as in Figure 1. Connect the lamp to a DC power supply together with an ammeter and voltmeter to record filament current and voltage respectively.
6. Place a radiation sensor (Thermopile) as shown in the bottom panel of Figure 1 and connect the radiation sensor to a digital multimeter that should be set on a 100 or 200-millivolt DC range.
7. For filament voltages between $1\,\mathrm{V}$ and $\sim 15\,\mathrm{V}$, in steps of about $0.2\,\mathrm{V}$, record the filament voltage (V), current (I) and the sensor millivoltmeter reading (V_{TC}).
8. Calculate the resistance of the filament $R\ (= V/I)$ for the various readings taken in step 7. Now calculate the temperature using the formula (6).
9. Draw a graph between $\log\,(V_{TC})$ and $\log(T)$ and estimate the slope.

Tables:

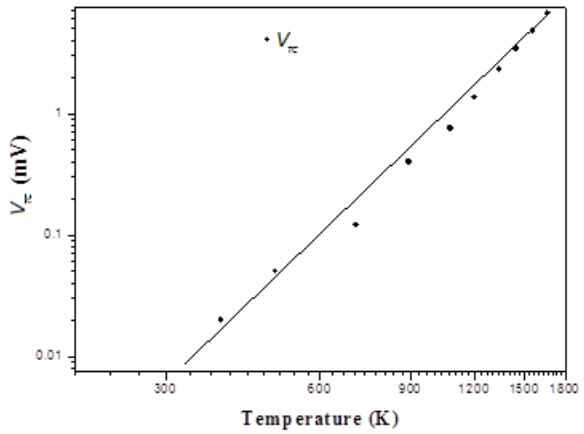
Table 1. Measurement of resistance (R) at low currents (less than 0.5 A)

S. No.	Current I (A)	Voltage (V)	Resistance R (Ω)

Table 2. Measurement of energy flux and temperature as a function of current (I_{max} = 2 A and V_{max} = 15 V)

S. No.	Current I (A)	Voltage (V)	Resistance R (Ω)	Energy flux V_{TC} (mV)	log(V_{TC})	Temperature T (K)	log(T)

Example:



Precautions:

1. Handle the thermopile only with the given rod.

2. Never touch the thermopile detector by hand.

3. Do not apply current more than 2.0 A through the filament.

3. Variable effective 'g' pendulum

Aim: (i) To determine (a) the acceleration due to gravity (g) by means of a compound pendulum and (b) its radius of gyration about an axis through its center of gravity.
(ii) To investigate the dependence of the value of 'g' on the plane of oscillations of the pendulum.

Apparatus: A 1 m long stainless steel rod as a compound pendulum with provision to tilt the plane of oscillations and a timer.

Theory:

A simple pendulum consists of a small body called a "bob" (usually a sphere) attached to the end of a string of negligible mass and of length large compared with the dimensions of the bob. Under these conditions the mass of the bob may be regarded as concentrated at its center of gravity, and the length of the pendulum is the distance of this point from the axis of suspension. When the dimensions of the suspended body are not negligible in comparison to the distance of the center of gravity from the axis of suspension, the pendulum is a compound, or physical pendulum. A rigid body mounted upon a horizontal axis so as to oscillate under the force of gravity is a compound pendulum.

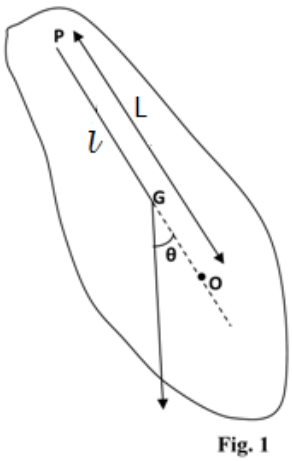


Fig. 1

Figure 1: A compound pendulum

In Figure 1, a body of irregular shape is pivoted about a horizontal frictionless axis through point P and is displaced from its equilibrium position by an angle θ. In the equilibrium position, the center of gravity G of the body is vertically below P. The distance GP is l and the mass of the body is m. The restoring torque for an angular displacement θ is

$$\tau = -mgl \sin\theta. \qquad (1)$$

For small amplitudes ($\theta \approx 0$),

$$I\frac{d^2\theta}{dt^2} = -mgl\,\theta, \qquad (2)$$

where I is the moment of inertia of the body through the axis P. Eq. (2) represents a simple harmonic motion and hence the period of oscillations is given by

$$T = 2\pi\sqrt{\frac{I}{mgl}}. \qquad (3)$$

By the parallel axis theorem, $I = I_G + ml^2$, where I_G is the moment of inertia of the body about an axis parallel to the axis through P and passing through the center of gravity G and

$$I_G = mK^2 \qquad (4)$$

where K is the radius of gyration about the axis passing through G.

Thus,

$$T = 2\pi\sqrt{\frac{mK^2+ml^2}{mgl}} = 2\pi\sqrt{\frac{\frac{K^2}{l}+l}{g}}. \qquad (5)$$

The period of a simple pendulum of length L is given by

$$T = 2\pi\sqrt{\frac{L}{g}}. \qquad (6)$$

Comparing Eqs. (5) and (6), the effective length of the compound pendulum is

$$\mathrm{L} = l + \frac{K^2}{l}. \qquad (7)$$

In other words, this is the length of an "equivalent simple pendulum": If the mass of the body were concentrated at a point O (See Fig.1) such that $\mathrm{OP} = l + \frac{K^2}{l}$, this would correspond to a simple pendulum with period T. The point O is called the "Centre of Oscillation".

Rearranging Eq. (7), we find

$$l^2 - lL + K^2 = 0. \qquad (8)$$

Equation (8) has two roots l_1 and l_2 such that

$$l_1 + l_2 = L \qquad (9)$$

and

$$l_1 l_2 = K^2. \qquad (10)$$

Thus, both l_1 and l_2 are positive. This means that on one side of the center of gravity (CG), there are two positions of suspension for which the periods are the same and there will be another pair of positions on the other side for which the periods are the same. Thus, there are four points of suspension, two on either side of the CG, about which the time periods of the pendulum would be the same. The distance between two such positions of the centers of suspension asymmetrically located on either side of CG is the length L of the equivalent simple

pendulum. Thus, if the body was supported on a parallel axis through the point O (see Fig. 1), it would oscillate with the same time period T as when supported at P.

From Eqs. (6) and (10), the value of g and K are given by

$$g = 4\pi^2 \frac{L}{T^2}, \tag{11}$$

$$K = \sqrt{l_1 l_2}. \tag{12}$$

By determining l_1, l_2 and L graphically for a particular value of T, the acceleration due to gravity g at that place and the radius of gyration K of the compound pendulum can be determined.

If the plane of oscillations is at an angle Φ with respect to the vertical plane (see Figure 2), the component of the acceleration due to gravity $g(\Phi)$ which is effective in its oscillation plane is reduced to $g(\Phi) = g \cos\Phi$ and Eq. (3) becomes

$$T(\Phi) = 2\pi\sqrt{\frac{I}{mgl\cos\Phi}} \tag{13}$$

In accordance with this formula, the increase in the oscillation period, which is proportional to the square root of $1/\cos\Phi$, can be confirmed experimentally.

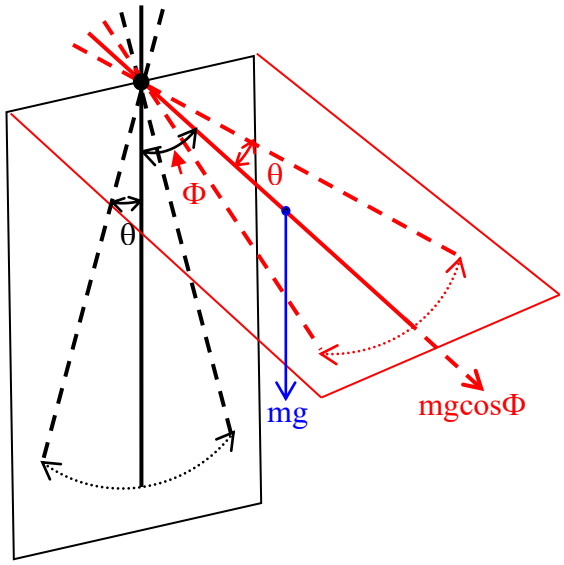


Figure 2: The physical pendulum set up for variable 'g' configuration.

Procedure:

In the present case, the pendulum is a rod of 10 mm diameter and 1 m length.

1. Clamp the rod on to the support stand using a clamping set up and suspend the rod in a vertical plane.
2. Allow the rod to oscillate in the vertical plane ($\Phi = 0$) with a small amplitude ($\theta < 10°$).
3. Measure the time taken for 10 oscillations using a precision stopwatch. Repeat this three times and find the mean time t for 10 oscillations and then determine the time period T. Give estimates for the error ΔT.
4. Measure the distance d of the free end of the rod from the axis of suspension using a meter scale.
5. Repeat steps 1 to 4 for different lengths (in steps of 2 cm) till the CG of the bar is approached, where the time period becomes very large (see Eqs. (6) and (7)).
6. Draw a graph with the distance $l = d - D/2$, where D is the total length of the rod, as x-axis and the time period T as y-axis. The graph will be as shown in Figure 3. (Note: l is the distance from the pivot point to the center of gravity of the rod. The center of gravity of the rod is located at D/2 for a straight cylindrical rod. We only need to plot one side of this graph, as we are not utilizing suspension points on the other side of the CG.)
7. Draw a line ABC parallel to the x-axis, as shown in Figure 3. The length L of the equivalent simple pendulum is $L = \mathrm{AC} + \mathrm{BC}$ and the radius of gyration is $K = \sqrt{\mathrm{AC} \times \mathrm{BC}}$. Also note down the time period T corresponding to the line ABC. Repeat the above for several lines parallel to the x-axis. For each line, obtain the values of L, T and K and estimate the value of g. Combine all your measurements to obtain a final estimate of g and Δg. Draw a graph with T^2 as x-axis and L as y-axis. Also draw a straight line with slope $g/4\pi^2$ for visual comparison.
8. For studying the dependence of g on Φ, repeat steps 1 to 3 for a length where T is minimum for values of Φ from $0°$ to $50°$. Plot $4\pi^2(K^2 + l^2)/l \cos\Phi$ vs T^2 for various Φ. Explain what you get and what you expect. (Use K obtained from step 7).

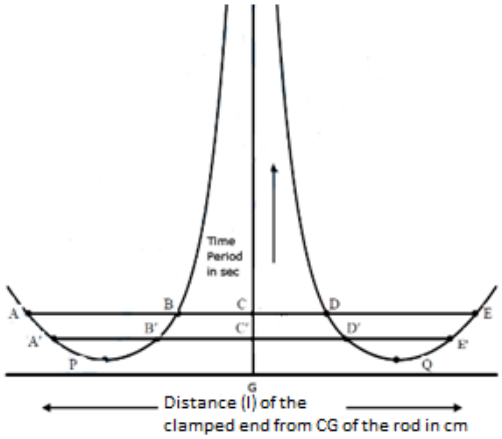


Figure 3: Time period vs. distance plot of the clamped end from CG of the rod

Table 1: Measurements for $\Phi = 0$

Serial no of holes from one end	Distance d of the hole from one end (cm)	Distance $l = d - D/2$ of the hole from CG (cm)	Time for 10 oscillations for three different trials (sec)			Mean time t for 10 oscillations (sec)	Time period T = t/10 (sec)

Table 2: To find the value of g and K

Sl.No.	Length $L = l_1 + l_2$ (cm)		T	T^2	$g = 4\pi^2(L/T^2)$	K
	l_1 (cm)	l_2 (cm)	(sec)	(sec^2)	(m/s^2)	(cm)
					Mean g =	Mean K =

Determine $K \pm \Delta K$ and from a graph of L vs T^2, find $g \pm \Delta g$.

Table 3: To find the dependence of g on Φ

Distance l for which measurements are taken: $l =$ _____ (cm)

Sl No.	Φ (deg)	Cos Φ	Time taken for 10 oscillations, three trials (sec)			Mean time t (sec)	Mean Time period $T = t/10$ (sec)	T^2 (sec^2)	$4\pi^2 (K^2 + l^2) / (l \cos\Phi)$

4. Transmission grating

Aim: Determination of the wavelength of the spectral lines of Mercury spectrum.

Theory: A grating is a transparent film or glass sheet on which a number of close-lying equidistant parallel lines are ruled. The ruled spaces are rendered opaque while the inter spaces remain transparent. In effect, therefore, the grating comprises of a large number of close-lying, parallel slits separated by equal distances.

Considering interference of parallel waves passing through slits, imagine a wave front is incident on the grating normal to the plane of the grating. Each slit becomes a secondary source of waves that spread out on the other side of the grating (Figure 1).

The path difference between the rays S_1A and S_2B from successive slits traveling in a particular direction, at an angle θ with the normal to the grating is $d\sin\theta$ where d is the separation of successive slits. If this path difference happens to be a multiple of the wavelength, when they arrive at a plane ABC, they will all be in phase and their amplitudes will all add up if their amplitudes are superposed. But they are all traveling in parallel direction and no superposition takes place unless the waves are converged to a common point. By placing a convex lens in the path of the beam, all the waves are converged at the focal point of the lens. As the paths AF, BF, CF etc. are optically equal and the waves are already in phase in the plane ABC, the wave reinforce one another at F and we get an image of maximum intensity. The amplitude is N times the amplitude of each wave and intensity will be N^2 times the intensity due to one slit, if there are on the whole N slits. The condition for obtaining maximum intensity is $dsin\theta = m\lambda$, where m is an integer. At these values, viz., $\theta = 0$, $\sin^{-1}\lambda/d$, $\sin^{-1}2\lambda/d$, etc., we will have maximum intensity in the image. These are called the principal maxima of orders 0,1,2, 3,…, etc.

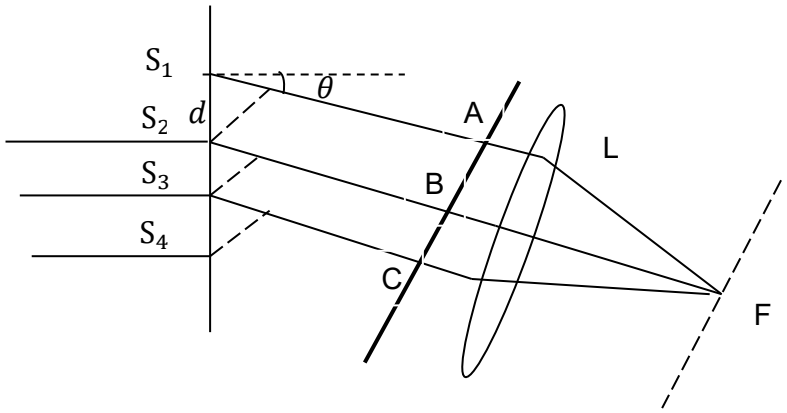


Figure 1

If the light that illuminates the grating is composite (consists of many wavelengths), then each order m has different angles. For each order, we will have a spectrum, and provided we know the distance between the rulings in the grating, we can determine the wavelength of the light from any source.

Experimental procedure:

To determine the wavelengths of the lines present in a given radiation, we measure, using a spectrometer, the angles at which the different orders (of maxima) are produced with respect to the incident beam. The grating can be standardized, i.e., the value of the grating element d can be determined using a standard monochromatic light source of known wavelength. Using this, the wavelength of any other source can be calculated.

Preliminary adjustments:

1. To adjust the incident beam from the collimator to be normal to the plane of the grating (Figure 1), the telescope is placed along the axis of the collimator as shown in Figure 2 and the image of the slit is obtained exactly at the intersection of the cross-wires (Figure 3). The reading in the circular scale corresponding to this position screw of the telescope is noted (R_1).

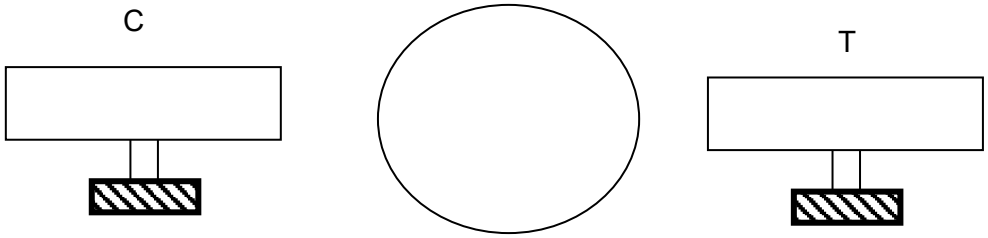


Figure 2

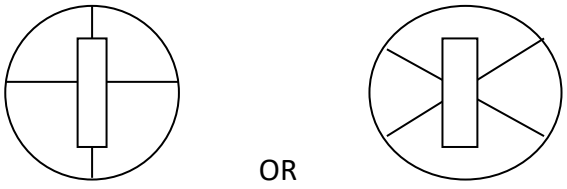


Figure 3

2. The fixing screw of the telescope is released and the telescope rotated to another position which reads (R_1 + 90°) or (R_1 − 90°), as shown in Figure 4. Then, the grating is mounted vertically on the central table. The central table is disengaged from the disc carrying the verniers and rotated till the image of the collimator slit after reflection at the grating surface is obtained exactly at the intersection of the cross wires. As the axes of the collimator and

the telescope have been set up to be perpendicular, this means that the beam from the collimator is incident on the grating at an angle of 45° (see Figure 5). The levelling screws may be adjusted if the image of the slit is not at the centre of the field of view.

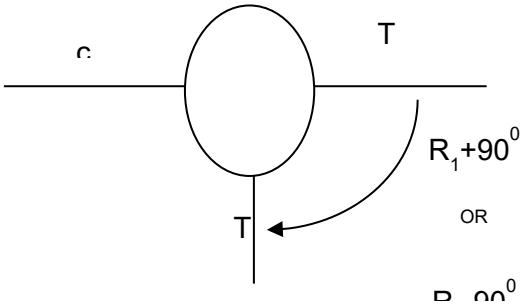


Figure 4

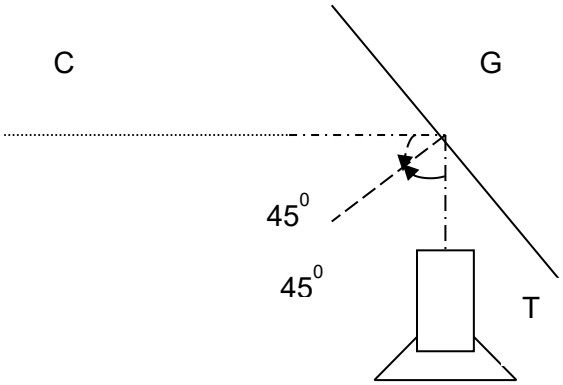


Figure 5

3. From the present reading, the central table is rotated exactly by 45^0 with the aid of the vernier scale in the proper direction, to make the grating normal to the direction of the incident beam from the collimator (Figure 6).
4. The slit of collimator is illuminated by the mercury vapour lamp. The telescope is moved to one side of the original direction and the angle with the direct beam is gradually increased, different colored lines are seen, constituting the spectrum of the mercury source (Figure 7).

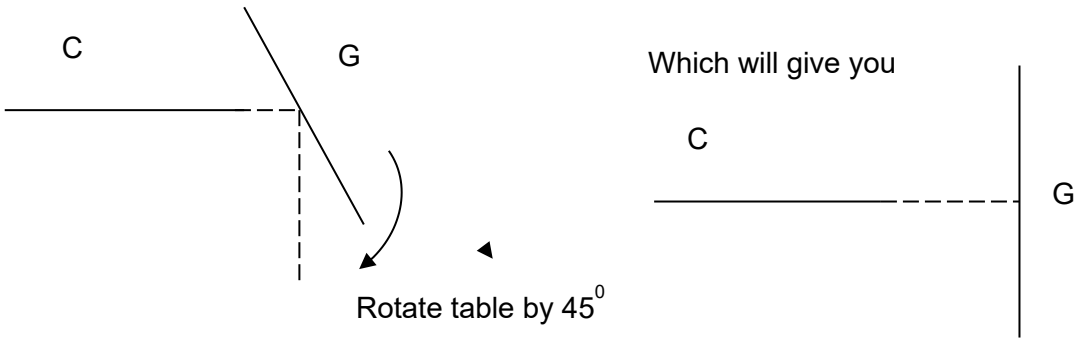


Figure 6

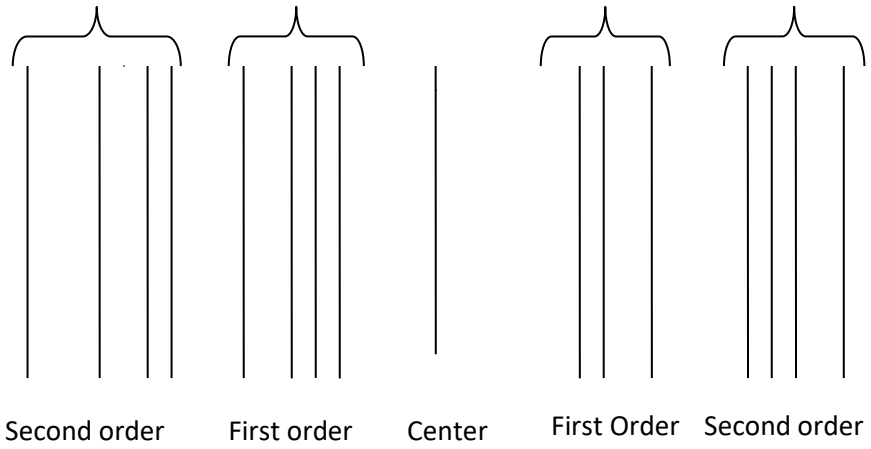


Figure 7

Readings:

1. The reading corresponding to the position of the telescope when it is in line with the collimator is again noted. This is bound to be different from the direct ray reading 'R' taken at the start, as the position of the vernier has been changed in the meantime.
2. Then the intersection of the cross wires is placed on the successive (prominent) lines of the first order spectrum and the reading (R) are taken for each line. The telescope is moved to the other side of the direct ray and the readings obtained for the same lines of the first order spectrum (R'). In the case of mercury spectrum, it consists of the following lines (Figure 8): Y_1, Y2, G and B are quite bright. The greenish-blue line is less intense. The two violet lines are of feeble intensity, the extreme violet line being relatively brighter one. It is a common mistake to assume the very bright blue line as the violet line. Once the violet lines are specially looked for and identified, the difference in shape can be appreciated. As noted in Figure 9, we see that (R-R') gives 2θ and hence θ can be found. **Note:** One of the wavelengths occurring in the mercury spectrum will be given to you. You can use this value to perform the standardization of the grating, i.e., determine the number of grating lines per unit length N, which is equivalent to knowing the grating spacing d.
3. The same experiment may be repeated with the second order spectra on either side.

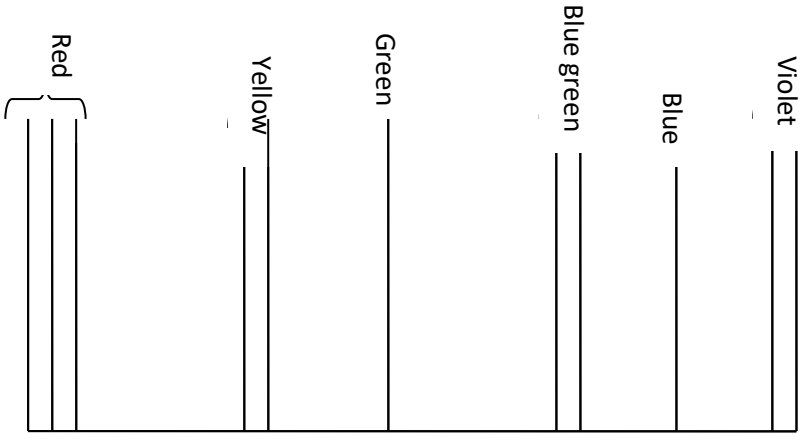


Figure 8

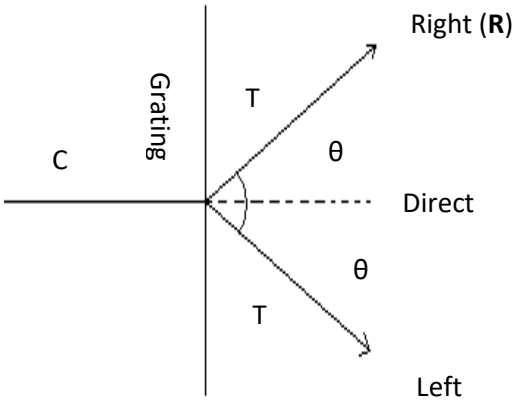


Figure 9

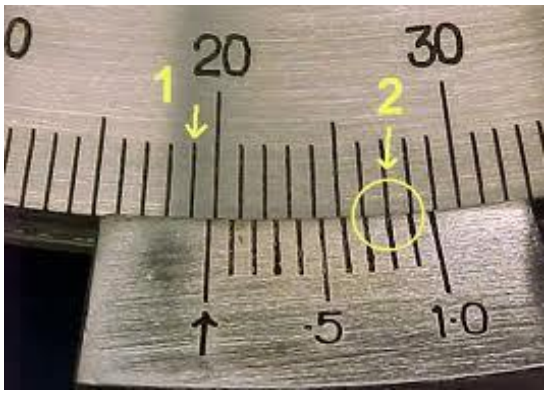


Figure 10

Reading a Circular Vernier Scale:

Main Scale is the fixed scale (at the top, in Figure 10) which has 0-360° marked on it. Vernier Scale is the small movable scale at the bottom.

Least Count (L.C.) = 1 M.S.D – 1 V.S.D

M.S.D = Main Scale Division

V.S.D = Vernier Scale Division

In this example:

1 M.S.D = 1^{O}

10 V.S.D's coincide with 9 M.S.D's.

Hence, 1 V.S.D = $\frac{9}{10}$ M.S.D

L.C. = 1 M.S.D – 0.9 M.S.D

L.C. = 0.1 M.S.D = 6'

Reading = Main Scale Reading + Vernier Scale Reading × L.C.

In this example:

Main Scale Reading (marked "1" in the figure 5 – the reading of the main scale just below the Vernier scale zero) = 19°

Vernier Scale Reading (the line on Vernier Scale coinciding with the Main Scale, shown as "2" in the figure 5) = 8

Reading = 19° + 8 x 6' = 19° 48'

Note that the least count of the spectrometer provided to you will be in the order of seconds. Note that 1°= 60' (minutes) and $1' = 60''$ (seconds).

1. Standardization of the grating

Known wavelength= _____ A° *Least count =*

Colour	Order (m)	Left (R)						Right (R ′)						R- R ′ = 2Θ (deg)	sinΘ	N = (sinΘ/mλ) lines/m
		Vernier A (Θ_A) (deg)			Vernier B (Θ_B) (deg)			Vernier A (Θ_A) (deg)			Vernier B ($\Theta_{B'}$) (deg)					
		Main Scale Reading	Vernier Scale Reading (VSCxLC)	Total Reading	Main Scale Reading	Vernier Scale Reading (VSCxLC)	Total Reading	Main Scale Reading	Vernier Scale Reading (VSCxLC)	Total Reading	Main Scale Reading	Vernier Scale Reading (VSCxLC)	Total Reading			

Vernier A (Θ_A) - Vernier A ($\Theta_{A'}$) = 2Θ

Vernier B (Θ_B) - Vernier B ($\Theta_{B'}$) = 2Θ

From the Equation, Sin Θ = N m λ, N = (Sin Θ /mλ) where N is the number of grating lines per meter.

2. Determination of other wavelengths (at least four different colours)

Colour	Order (m)	Left (R)						Right (R ′)						R- R ′ = 2Θ (deg)	sinΘ	λ = sinΘ/mN (A°)
		Vernier A (Θ_A) (deg)			Vernier B (Θ_B) (deg)			Vernier A ($\Theta_{A'}$) (deg)			Vernier B ($\Theta_{B'}$) (deg)					
		Main Scale Reading	Vernier Scale Reading (VSCxLC)	Total Reading	Main Scale Reading	Vernier Scale Reading (VSCxLC)	Total Reading	Main Scale Reading	Vernier Scale Reading (VSCxLC)	Total Reading	Main Scale Reading	Vernier Scale Reading (VSCxLC)	Total Reading			

Summarize your results.

5. Mapping equipotential lines

Aim: To map regions of equipotential for a given set of electrode configurations.

Apparatus: A circular glass container with KCl solution (0.1 g of KCl in 250 ml distilled water should be prepared), two pairs of electrodes with (i) circular and (ii) parallel plate configuration, ac voltage source, multimeter and connecting wires.

Introduction

A simple way to understand the force of interaction between electric charges is to consider that a charge produces a field in the space surrounding it. Any other charge kept in this field would experience a force, the Coulomb force. Hence, the action at a distance can be thought of as being facilitated by the electric field. The electric field is defined as the force per unit charge exerted on a test positive charge. An associated concept is the potential energy, which is similar to the potential energy of a mass in a gravitational field. The electric potential energy at a given point is defined as the work done in bringing a test charge from infinity up to that point. The work done on a unit positive charge is the electrostatic potential. Since the potential energy is defined with respect to a convenient but arbitrary reference point where it is taken to be zero (a point at infinity here), the absolute potential is not a measurable quantity. Instead, it is the potential difference between two points that has observable consequences and is a physical quantity that can be measured. For instance, in a common 1.5 V dry cell the positive terminal is at a potential of 1.5 V higher than its negative terminal. Note carefully the use of potential (measured in units of Volts) and not potential energy. The primary difference is that potential is independent of the magnitude of the test charge, just as the gravitational potential does not depend on the magnitude of the test mass.

Another related concept is regions of constant potential or equipotential regions. The potential difference between any two points in an equipotential region is zero. When a charge is moved along an equipotential region, the change in potential energy of the charge is zero which means the work done by the field on the charge is zero. Since the electric field points in the direction of decreasing potential, the lines of electric field and equipotential lines form a set of orthogonal curves. In three dimensions, regions of constant potential are surfaces and are termed equipotential surfaces. In our experiment, equipotential points can be identified by connecting the two probes of a voltmeter (multimeter set in voltage mode) across any two points and if the meter reads 0V, then both points are at the same potential.

Experimental Procedure

A Photograph of the experimental arrangement is displayed in Figure 1, while a schematic of the electrode configuration is shown in Figure 2. To begin with use, the circular configuration. The goal is to locate a region, all the points of which are at constant potential with respect to one of the electrodes.

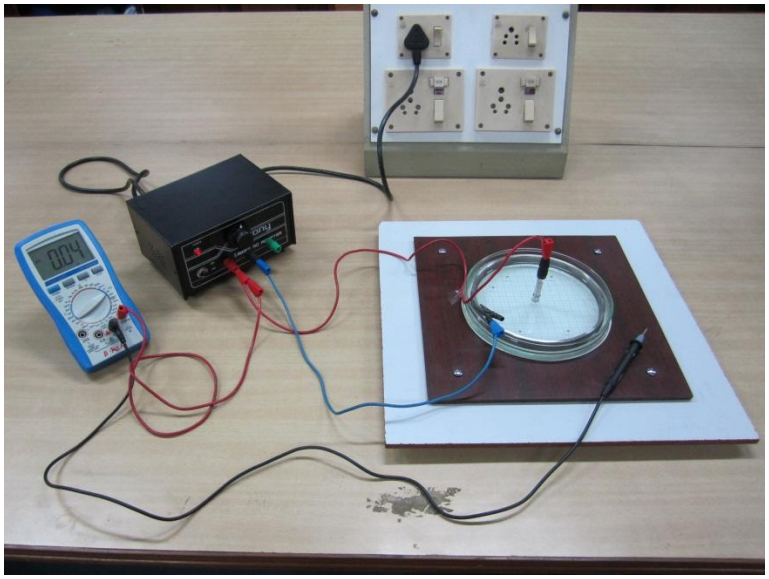


Figure 1: Experimental set up for mapping equipotential lines

Procedure

1. Apply a suitable voltage of about 6 V (rms) between the inner electrode and the outer annular electrode which is kept concentric with the inner electrode. The outer electrode must be at ground potential.
2. Set the multimeter to read **ac voltage**. Note one of the probes (black) of the multimeter is insulated completely but for a small portion at the tip of the probe. This probe should be used for measurements, while the other probe (red) should be connected to the inner electrode.
3. Immerse the "black probe" in the solution until it comes into contact with the bottom and move it until the voltmeter reads 2.5V. Essentially, this tells you that the voltage difference between inner electrode and the point where the probe is located is 2.5V.
4. Read the coordinates of the point from the graph sheet placed at the bottom and transfer the same to your graph sheet. Prior to this, two concentric circles must be drawn on the graph sheet with the radius of the inner circle taken to be the radius of the inner electrode and the radius of the outer circle taken to be the inner radius of the outer electrode.
5. Repeat the experiment by locating several points that are at the same potential of 2.5V with respect to the outer electrode. Take as many points as possible so that you can connect all the points by a meaningful smooth curve. This curve gives you an equipotential region.
6. Repeat the above for the voltage differences in steps of 1 V until 6.5 V.
7. Once the experiments are completed, take out both the electrodes and place the two parallel electrodes in the solution. See the right panel of Figure 3 for the placement of the electrodes. There are two protrusions below the electrodes and they must fit inside the two

slots on the base. This will provide mechanical stability. Apply a potential of about 6 V between the two electrodes and repeat the steps 2-6 for this electrode configuration.

8. Discuss your observations in the context of the theory provided below and complete the activities described in the theory section.

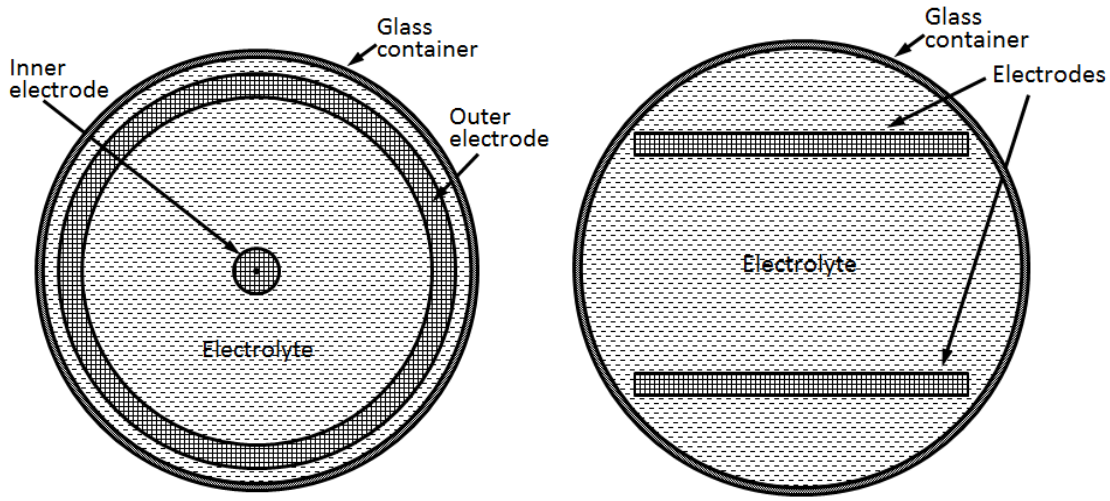


Figure 2: Electrode placement for circular (left) and parallel plate (right) configurations

Theory

Equipotential lines and electric field for cylindrical configuration: The potential distribution in a given region can be obtained by solving the Laplace's equation under suitable boundary conditions. The circular symmetry of the electrodes used in the first experiment implies that the Laplace's equation be considered in polar coordinates. In these coordinates and with the understanding that there is no angular dependence, the Laplace's equation becomes

$$\frac{1}{\rho}\frac{d}{d\rho}\left(\rho\frac{dV}{d\rho}\right) = 0 \qquad (1)$$

Excluding the $\rho = 0$, from the solution and integrating Eq. (1) twice yields

$$V = A\ln\rho + B \qquad (2)$$

where, A and B are constants of integration. Let the outer electrode be at the potential V_0 and the inner electrode be at the potential 0. The boundary conditions can be written as

$$V(a) = V_0, \text{ and } V(b) = 0, \qquad (3)$$

where, a is the radius of the inner electrode and b is the inner radius of the outer electrode. Note here that $b > a$. Substitution of the boundary conditions yields the solution for the potential at any point in between the electrodes as:

$$V(\rho) = V_0\frac{ln(b/\rho)}{ln(b/a)}. \qquad (4)$$

Note that the solutions correctly predict the potential difference between the electrodes.

Setting V to be constant (the LHS) lead to the equation that gives the equipotential curves as

$$\rho = \text{constant}, \qquad (5)$$

which are circles of constant radii.

The electric field can be found as $\vec{E} = -\vec{\nabla}V$. Note that the gradient has to be taken in cylindrical coordinate. This leads to

$$\vec{E}(\rho) = \frac{V_0}{\rho}\frac{1}{ln(b/a)}\hat{e}_\rho \qquad (6)$$

The electric field is along the radial direction and its magnitude decreases with radial distance. Electric field lines are defined as curves, the tangent to which at any point gives the direction of the electric field at that point. This definition can be used to get the equation for electric field lines. Figure 3 shows a sketch of equipotential lines and the electric field lines.

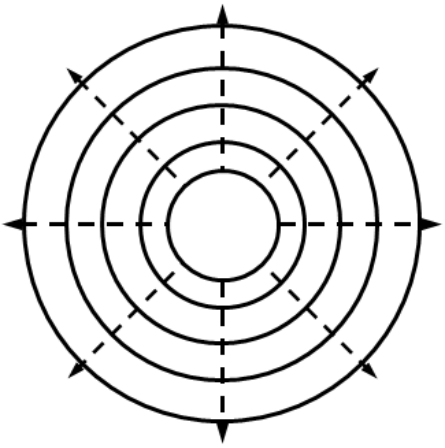


Figure 4: Equipotentials (circles) and electric field lines (dashed radial lines) for the electrode geometry considered.

Draw a few radial lines in the graph sheet. You may draw dashed lines to differentiate them from equipotential lines. Also use Eq. (4) to get equation for the equipotential for at least one circle.

Uniqueness Theorem: It is important at this stage to know the uniqueness theorem. In a rather simplified form, it can be stated that given the boundary conditions, the potential at any point in between closed boundaries is uniquely and completely determined by the Laplace equation and is independent of the potential outside the boundary. In other words, given the potentials on the electrodes (boundaries) the potential at any point in between the electrodes can be uniquely determined.

Equipotential lines and electric field for parallel configuration: The equation for equipotential lines can also be derived for the parallel configuration. Assuming the potential to be function of *y* coordinate alone, the Laplace's equation can be written as

$$\frac{d^2V}{dy^2} = 0. \qquad (7)$$

Replacing the partial derivative with total derivative and integrating it twice yields

$$V = Cy + D, \qquad (8)$$

where C and D are constants of integration. Assuming the boundary conditions,

$$V(d) = V_0 \text{ and } V(0) = 0, \qquad (9)$$

where d is the distance between the electrodes, leads to

$$V(y) = V_0 \frac{y}{d}. \qquad (10)$$

Thus $y = \text{constant}$ (horizontal lines) are the equations of the equipotential lines. The electric field this case can be found as $\vec{E} = -\vec{\nabla}V$

$$\vec{E}(y) = -\frac{V_0}{d}\hat{e}_y. \qquad (11)$$

The electric field is constant and points along the negative y axis, while the electric field lines are vertical lines ($x = \text{constant}$). See Figure 5 for equipotentials and electric field lines for the parallel plate configuration.

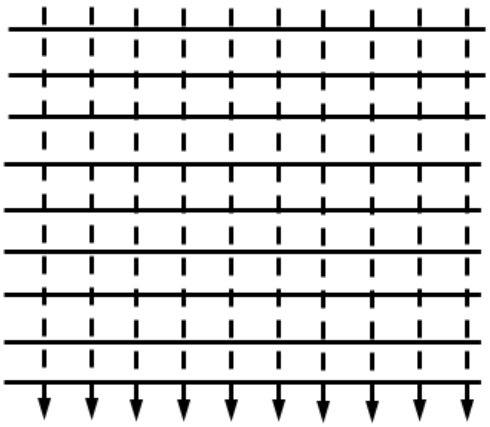


Fig.5 Equipotentials (continuous horizontal lines) and electric field lines (dashed vertical lines) for the parallel plate electrode geometry.

Sketch a few vertical lines in the graph sheet. You may draw dashed lines to differentiate them from equipotential lines. Also use Eq. (11) to get the equation for the equipotential line for at least one case.

You may have noticed that the experimentally mapped equipotential curves are not straight lines as predicted by the theory. Try to find the reason for this.

6. Ultrasonic Interferometer – Velocity of ultrasonic waves

Aim: To determine the velocity of ultrasonic waves using ultrasonic interferometer.

Apparatus: Ultrasonic interferometer setup (High frequency Generator 2 MHz, measuring cell, circular base to hold the cell, coaxial cable), distilled water, beaker, interferometric setup pre-filled with glycerine or gylcerine solution.

Theory:

Sound waves are characterized based on their frequency:

1. Audio waves: Frequency lies between 20 Hz to 20000 Hz
2. Infrasonic waves: Less than 20 Hz
3. Ultrasonic waves: Higher than 20000 Hz

An ultrasonic interferometer offers a precise and reliable method for determining the velocity of ultrasonic waves in liquid media. The measurements obtained from this device are highly accurate and consistent. For this study, a crystal-controlled interferometer operating within a frequency range of 1 to 12 MHz is employed to measure the ultrasonic velocity. Ultrasonic waves are produced by the piezoelectric effect. The principle used in the measurement of velocity (v) is based on the accurate determination of the wavelength (λ) in the medium. Ultrasonic waves of known frequency are produced by a quartz crystal which is fixed at the bottom of the cell. These waves are reflected by a movable metallic plate kept parallel to the quartz crystal. Standing waves are formed whenever the distance between the two plates l is such that

$$l = \left(n + \frac{1}{2}\right)\frac{\lambda}{2}.$$

To see this, note that the bottom of the cell where the source, i.e. the quartz crystal, is located corresponds to an anti-node for the sound wave, since the wave has maximum amplitude at the source. The upper end of the cell, i.e., the movable metallic plate, constitutes a node for the wave where its amplitude vanishes. When the cell length satisfies this condition, an acoustic resonance occurs that gives rise to an electrical reaction on the generator driving the quartz crystal, and the anode current of the generator becomes maximum. Thus, if the cell length is increased or decreased by exactly a half wavelength ($\lambda/2$) or multiples of it, anode current

becomes maximum. If d is the separation between successive adjacent maxima of anode current, then we can see that

$$d = \lambda/2.$$

The velocity (v) of a wave is related to its wavelength (λ) by the relation

$$v = f\lambda.$$

The velocity of ultrasound is determined principally by the compressibility of the material of the medium. For a medium with high compressibility, the velocity will be less. Adiabatic compressibility of a fluid is a measure of the relative volume change of the fluid as a response to a pressure change. Compressibility is the reciprocal of bulk modulus and is usually denoted by β. The adiabatic compressibility of the material of the sample can be calculated using the equation,

$$\beta = \frac{1}{\rho v^2},$$

where ρ is the density of the material of the medium and v is the velocity of the sound wave through that medium.

Ultrasonic Generator:

Ultrasonic waves can be generated through two primary mechanisms: Mechanical generators and electrical generators. Electrical generators are further categorized into piezoelectric generators and magnetostriction generators. In this experiment, we use a piezoelectric generator. The ultrasonic generator generates a high frequency alternating current (AC) signal that drives the transducer, which is a quartz crystal. The generator contains an oscillator circuit that produces a stable frequency signal (in our case it is 2 MHz). This signal is then amplified to a suitable level to drive the ultrasonic transducer. The ultrasonic transducer, which is fixed at the bottom of the measuring cell and usually made of piezoelectric material, vibrates at the frequency of the electrical signal, creating ultrasonic waves. These waves are transmitted into the medium in which we are interested to measure the speed of sound.

Measuring cell:

The measuring cell is a specially designed double walled brass cell with chromium plated surfaces having a capacity of 10 ml. The double wall allows water circulation around the

experimental medium to maintain constant temperature during the experiment. A fine micrometer screw (25 mm length and 0.01 mm least count) is provided at the top of the cell which helps to raise or lower the reflector plate in the cell through a known distance. It has a quartz crystal fixed at the bottom.

A photo of the experimental setup and a schematic of the measuring cell is shown in Figure 1.

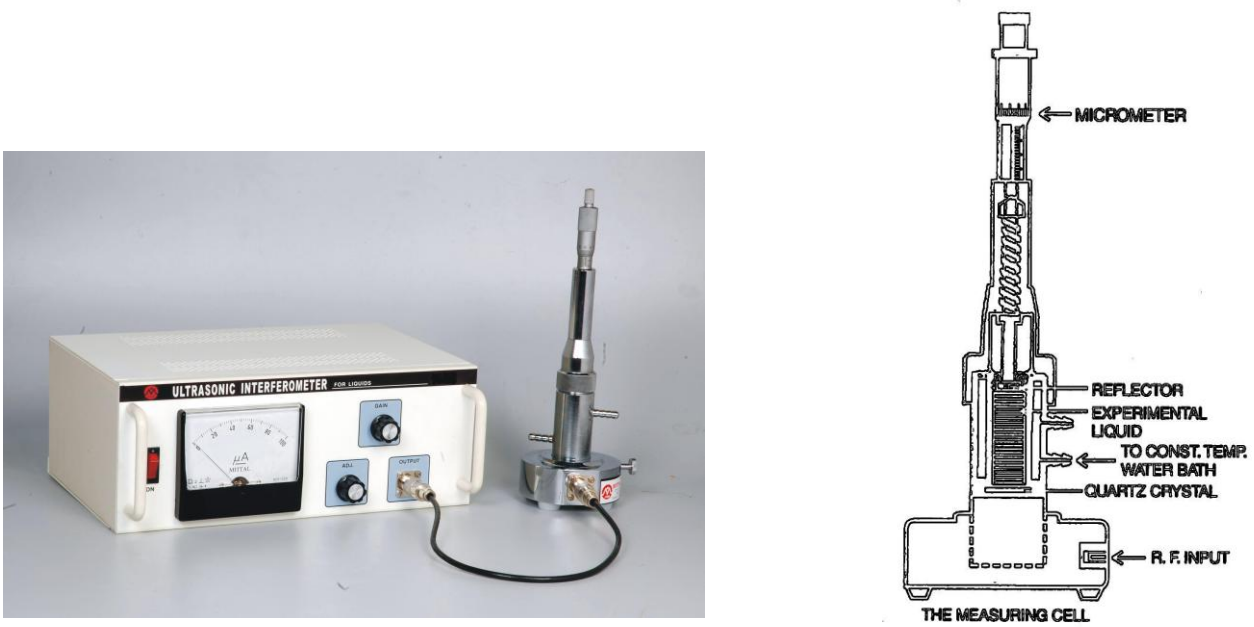


Figure 1: Ultrasonic interferometer setup (left) and schematic of the measuring cell

Experimental Procedure:

1. Fill the interferometer volume with water and make sure there are no air pockets.

2. Keep the "**ADJ**" knob at the centre approximately and "**GAIN**" Knob at maximum.

3. Switch on the unit and wait for some time. If the needle position goes below or above the scale, it can be brought to the scale by using the **"ADJ"** knob.

4. The micrometer can be rotated in clockwise or anti-clockwise direction but the maximum and minimum current values should lie within the range of the scale, i.e., $0 - 100$ μA.

5. Note the readings of the micrometer corresponding to the n number of maximum (or minimum) values. It is preferred to record every third maxima or minima in order to reduce measurement error (see table below). Calculate the distance d, which is the mean shift between successive maxima (or minima), using the recorded data.

6. From the mean shift, estimate the wavelength of the ultrasonic waves and thereby determine the velocity of the waves. Provide an error estimate for both the quantities.

7. Use the determine velocity to compute the adiabatic compressibility β and estimate the uncertainty.

8. Repeat the experiment with glycerine or glycerine solution.

Table 1: To find the ultrasonic velocity of waves in the given liquid.

Least count of the micrometer:

Reflector set on maxima	Micrometer readings for maximum (or minimum) deflections			Shift for $(n+15)-n$ 10^{-3} mm	Mean Shift 10^{-3} mm
	Pitch Scale Reading	Head Scale Reading	Total Reading (PSR+HSR) 10^{-3} mm		
n					
$n+3$					
$n+6$					
$n+9$					
$n+12$					
$n+15$					
$n+18$					
$n+21$					
$n+24$					

Precautions:

- Liquid should be filled in the cell when it is detached from the circular base and wipe out the excess the liquid overflowing from the cell.
- Do not switch on the unit without filling the liquid in the cell.
- Do not tilt the cell after filling the liquid to avoid flow of liquid towards micrometer which may rust/jam the threads of the micrometer head.
- Remove the experimental liquid from the cell after completion of experiment.

- Keep micrometer open at 25 mm after use.
- Wait for some time, say 15 seconds for ultrasonic interferometer set up to warm up before taking readings.

7. Wavelength of light by interference - Newton's rings

Aim: To determine the wavelength of a monochromatic light by forming Newton's rings.

Apparatus: A plano-convex lens of large focal length, glass plate, monochromatic light source (sodium vapor lamp), a traveling microscope.

Theory:

Interference

Interference effects are observed when, in a certain region of space, two waves of the same frequency are superposed and the two waves reach the same point with some phase difference. Depending on the phase difference at any point, the effect of superposition is to produce varying intensities ranging from a maximum of $(A_1 + A_2)^2$ to a minimum of $(A_1 - A_2)^2$, where A_1 and A_2 are the amplitudes of the individual waves. For the interference pattern to be observed, the intensities at different points and hence the phase differences between the individual waves should be constant in time. Such waves having a phase relation which is independent of time are said to be coherent. The light waves from two different sources typically do not have a stable phase relationship. So, it is necessary that the two waves that are made to travel different paths and subsequently superposed to produce interference, are derived from a single source wavefront. The splitting of a single wavefront into two portions to produce interference may be achieved in two ways.

(a) By wavefront division as is done in a double slit experiment.

(b) By making the rays undergo partial reflection and interfering the reflected/refracted waves. An example is the production of interference bands by partial reflection at the two faces of a thin transparent film as shown in Figure 1.

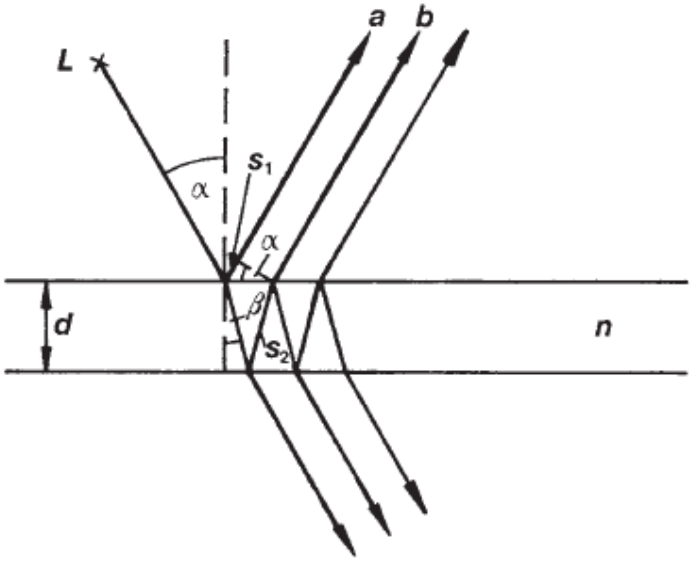


Figure 1: Production of inteference bands by partial refelection at the two faces of a thin transparent film

In Figure 1, compared to ray a, ray b has gone through a longer path and therefore has a phase lag. It can be proved from the geometry that the net phase difference between the wavefronts corresponding to rays a and b is

$$\delta = \left(\frac{2\pi}{\lambda}\right) 2nd\cos\beta + \pi.$$

Depending on the value of δ, there will be a maximum, minimum or intermediate intensity along the line of observation of the reflected light. The second term is because depending on the actual film, one of the reflections (either for ray a or b) typically occurs as the light is incident from a rarer medium (lower refractive index) on to a denser medium (higher refractive index), which leads to an extra phase of π in the reflected light.

A variation in the phase difference δ can arise due to a change in the separation d. If our thin film consists of an air wedge between two glass plates with varying thickness d, we obtain a series of parallel equidistant bright and dark bands. The rays reflected by the two bounding faces of the air film intersect very close to the surface of the glass plate and the interference bands are formed close to the air wedge. By focussing a microscope on the surface of the glass plate, one would be able to view the pattern. Such fringe systems formed close to the air wedge arising mainly due to the variation in the thickness of the film are called fringes of equal thickness, because each fringe of a given intensity corresponds to the locus points having the same value of thickness d.

Newton's Rings:

Referring to Figure 2, a convex lens is placed over a plane glass plate, forming an air wedge of varying thickness. The system is illuminated from above by a monochromatic light from a sodium lamp. Interference fringes are formed due to the superposition of light reflection by the upper and lower boundaries of the air film. Loci of constant d will be concentric circles around the point of contact of the lens with the plate. A band of minimum intensity is obtained if the separation satisfies the relation $(2\pi/\lambda)2nd\cos\beta + \pi = (2m+1)\pi$. Assuming near-normal incidence, $\cos\beta \approx 1$, and since $n = 1$ for air, this condition yields $2d = m\lambda$ or $d = m\lambda/2$. The center correspoonds to $d = 0$ and this condition is satisfied with $m = 0$. At a small distance away from the center, where the air thickness is half a wavelength, we have the first order dark ring. When $d = m\lambda/2$, we have the mth order dark ring. Midway between successive dark rings, we have rings of maximum intensity corresponding to $d = (2m-1)\lambda/4$.

To obtain the radii of the different rings, we must relate the separation d to the distance from the centre. In the right panel of Figure 2, if r is the radius of the ring and R the radius of curvature of the convex surface, we have

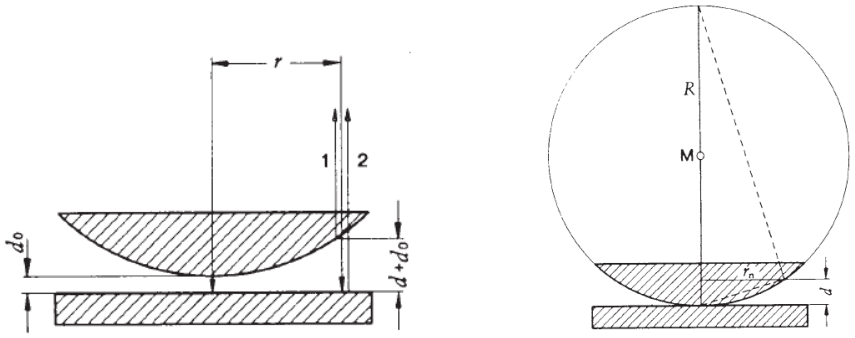


Figure 2: A plano-convex lens placed over a glass slab, forming an air wedge of varying thickness d (left panel) and geometrical relation between r, R and d (right).

$$r^2 = R^2 - (R-d)^2 \text{ or } d = r^2/2R.$$

Hence, for dark rings we find the condition

$$r_m{}^2 = m\lambda R,$$

while for the bright rings, we get

$$r_m{}^2 = (2m-1)\lambda R/2\,.$$

Measuring the diameters of different orders of the dark rings in the pattern, we may find λ if R is known.

Experiment:

Note: The convex lens should have focal length of 75 cm or larger.

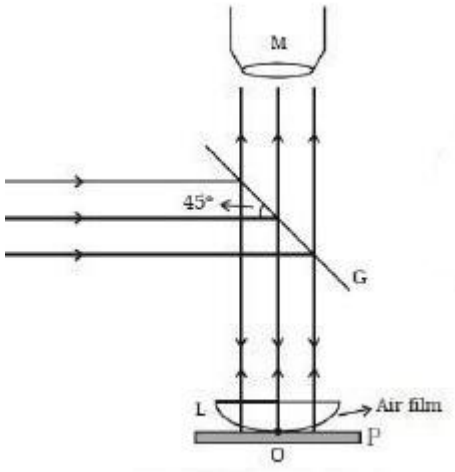


Figure 3: Experimental schematic of Newton's rings experiment

1. Place the biconvex or plano-convex lens over a plane glass plate. It will be better if the lower portion of the glass plate is a ground surface to avoid reflection from the lower surface.
2. Mount a transparent glass plate above the center of the lens at about 45° to the horizontal (see Figure 3)
3. Mount the whole arrangement on the horizontal platform of a vernier microscope.
4. Keep the microscope along with the lens system on it facing the sodium vapor lamp. When the lens is viewed from above, through the 45° plate, the reflected image of the lamp will be seen at the centre of the lens. (A slight change in the orientation of the microscope base with respect to the lamp may be necessary to get the bright patch of light over the centre of the lens). Even with the naked eye, a tiny dark spot (a small ring system) can be seen inside the bright patch.
5. If the same is viewed through the microscope/camera, a number of alternately bright and dark concentric rings can be seen (see Figure 4).
6. Measure the diameter for different rings and find out the wavelength of the given source of light. The measurements are made using the Newton's ring software. Please refer to the procedure described below to use the software.
7. Using the data you have tabulated in Table 1 (see below), estimate the wavelength of the light using the equations discussed in the theory. Estimate the uncertainty in your measured value. Draw a graph with order m on x-axis and the (diameter)^2 along y-axis. Also draw a straight line with appropriate slope (using the equations in the theory) and your estimate of the wavelength.



Figure 4: Newton's rings as observed through a microscope/camera

Newton's rings experiment software

Procedure:

1. Switch on the source. Wait for the yellow colour light.

2. Open the Newton's ring software on the computer monitor.

3. Select start streaming. Now arrange and adjust the plane glass, lens setup to get the Newton's rings with best contrast. Verify with your teacher.

4. Now click on the "start streaming" key. Observe the green grid.

5. Adjust the dark and bright concentric circular fringes on the screen in such a way that the horizontal line passes through the center.

6. Now Press the “capture” key to capture the image.

7. The captured image can be zoomed-in or zoomed-out by pressing the shift '+' or shift '-' keys.

8. Press the center calibration key. A message “***Center calibration mode activated***" will appear on screen. Click 'Ok'.

9. Press “Label ring” key, then “***Labelling mode activated***.” message will appear on screen. Click on the ring to label them.

10. Select the circular fringe number and mark the intersection points of the circular fringe and center horizontal line on both sides.

11. (***If you want to clear the markings press start streaming. And do the above steps once again from step 4.)***

12. Similarly mark as many circular fringes as you wish to measure their diameter.

13. Now click measure Newton’s ring.

14. Click over the labels then find the distance (1,1), (2,2), (3,3) and so on and look for the diameters of the rings on the screen. It gives the average diameters of our individual measurements. Tabulate your readings for further calculations (take minimum 5-6 readings for each ring).

15. After noting down the average diameter, click clear the table for the new measurement readings.

16. Save your data and image on the computer with your Roll number. (ex:ME24Bxxx)

Table 1: To find the diameter of the rings.

	Readings from software		Diameter of the ring (mm)	Uncertainty in Diameter (mm)	Diameter square of the ring $(mm)^2$
	Left co coordinate	Right coordinate			

Focal length of the given plano convex lens: _____ cm

8. Transistor Characteristics

Aim: To study the static characteristics of a Transistor in common emitter configuration.

Apparatus required: NPN transistor, power supply, voltmeter, ammeter.

Theory:

A transistor is a solid-state device used to amplify or switch electronic signals and electric power. A transistor consists of two P-N junctions formed by sandwiching either a P-type or N-type semiconductor between a pair of the opposite type. There are two types of transistors, namely, N-P-N transistors and P-N-P transistors, see Figure 1. The transistor can be regarded as a combination of two diodes connected back-to-back. There are three terminals, one taken from each type of semiconductor. The emitter, base and collector are provided with terminals which are labeled as E, B and C. The circuit symbols employed for P-N-P and N-P-N transistors are as shown in Figure 1. The arrow head is always at the emitter (not at the collector) and in each case, its direction indicates conventional direction of current flow. For a PNP transistor, the arrow head points from emitter to base (emitter is positive with respect to base and with respect to the collector). For an NPN transistor, the arrow head points from base to emitter (base is positive with respect to emitter and the collector).

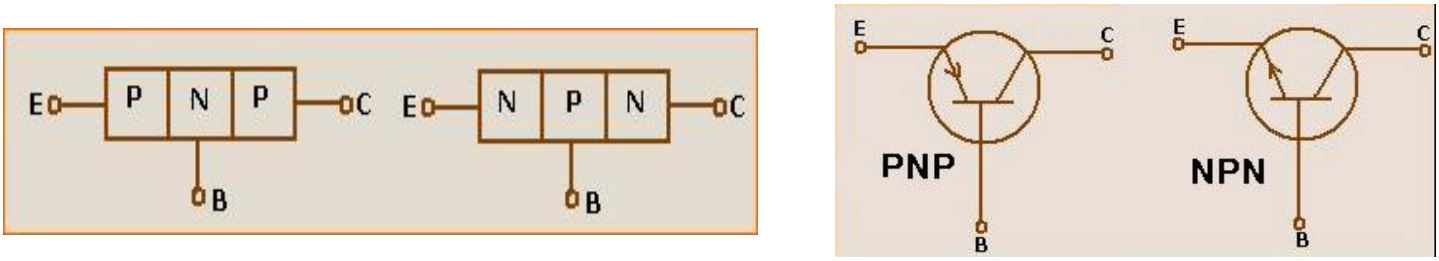


Figure 1: Schematic of a P-N-P and N-P-N type transistor (left) and their circuit symbols (right).

1. **Emitter**: It forms the left-hand section of the transistor. It is more heavily doped than any other region because its main function is to supply majority charge carriers, either electrons or holes. (electrons if emitter is N-type and holes if emitter is P-type).
2. **Base**: It forms the middle section of the transistor. It is very thin (of the order of 10^{-6} m) when compared with other sections, emitter and collector, and is very lightly doped.
3. **Collector**: It forms the right-hand section of the transistor and is lightly doped. Its main function as the name implies is to collect the majority charge carriers through base. The collector region is made physically larger when compared with other regions because it has to dissipate much greater power. And because of this reason, we cannot interchange the emitter and collector regions. However, for the sake of convenience, it is customary to show emitter and collector to be of equal size.

There are 3 leads in a transistor viz emitter, base and collector terminals. However, when a transistor is to be connected in a circuit, we require four terminals, two for input and two for the output. It can be done by making one transistor terminal as common to both input and output terminals. The input is fed between this common terminal and one of the other two terminals. Accordingly, a transistor can be connected in a circuit in three different ways. Each circuit has its own advantages and disadvantages.

i. Common base configuration (CB)
ii. Common emitter configuration (CE)
iii. Common collector configuration (CC)

For proper working of a transistor, it is essential to apply voltages of correct polarity across its two junctions. A PN junction is said to be forward biased if the P side is at a higher applied voltage than the N side, and is said to be reverse biased if the N side is a higher applied voltage than the P side. Based on the polarity of voltages applied across the two junctions, there are three modes of operation:

i. Cutoff mode: both junctions are reverse biased.
ii. Saturation mode: both junctions are forward biased
iii. Active mode: base - emitter junction is forward biased and collector –base is reverse biased.

In active mode, the forward-biased emitter–base junction allows majority charge carriers (electrons) from the emitter to diffuse into the base. Since the base is thin and lightly doped, most of these electrons do not recombine and instead reach the collector–base junction. The reverse-biased collector–base junction then sweeps these minority carriers (electrons) in the base (which, recall, is doped with holes) into the collector by its electric field, resulting in a net flow of electrons from emitter to collector.

The fundamental relation between the currents in a transistor circuit can be obtained by Kirchhoff's current law and is given by,

$$I_E = I_B + I_c,$$

which is true for all types of transistors and all transistor configurations.

Common Emitter Static Characteristics:

The static characteristics of a transistor in CE configuration are determined by the circuit shown in Figure 1. A micro-ammeter is connected in series with the base to measure I_B, the base current. Similarly, a milli-ammeter is connected in series with the collector circuit to measure I_C, the collector current. A voltmeter is connected across base and emitter terminals for measuring V_{BE} and a second voltmeter is connected across collector-emitter terminals to measure the output collector–emitter voltage, V_{CE}. The rheostats R_2 and R_1 are used to vary the base voltage (input voltage) and the collector voltage (output voltage).

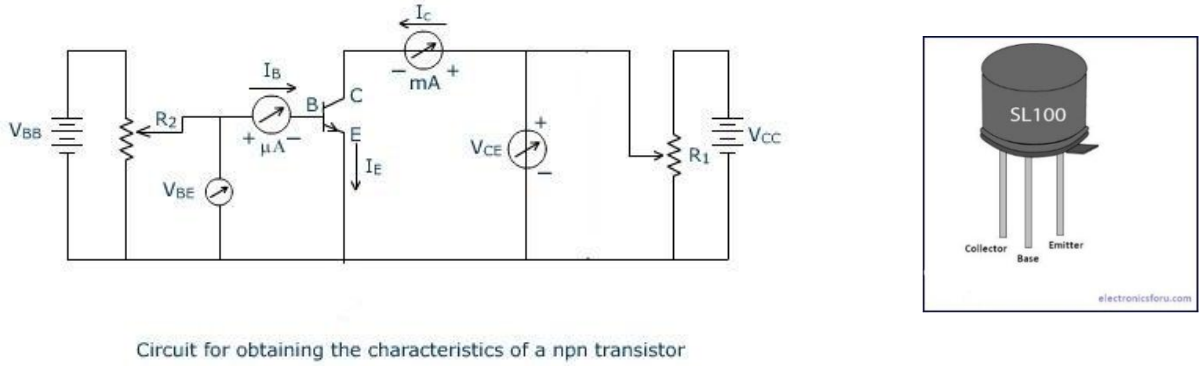


Figure 1: Circuit diagram for determining transistor characteristics in common emitter configuration (left) and diagram show the three pins of a transistor (right).

Procedure:

(I) Input Characteristics: It shows how base current, I_B varies with changes in V_{BE} when V_{CE} is kept constant.

1. Switch on the unit. Initially keep the knob R_2 to minimum position and adjust the collector voltage V_{CE} to 5 V by R_1 and keep it constant for the whole experiment.

2. Increase the base - emitter voltage V_{BE} in steps, and each time measure I_B.

3. Plot I_B (in μA) vs. V_{BE} (in volts).

4. Repeat the above experiment for different values of V_{CE}.

The overall shape resembles the forward characteristics of a PN diode as shown in Figure 2. The reciprocal of the slope gives the input impedance R_{in} of the transistor, which is defined as the ratio of small change in base-emitter voltage to the corresponding change in the base current at a given V_{CE}:

$$R_{in} = \frac{\Delta V_{BE}}{\Delta I_B}\bigg|_{V_{CE}=\text{constant}}.$$

Due to initial non-linearity of the curve, R_{in} varies considerably from a value of 4 kΩ near the origin to a value of 600 Ω over the more linear part of the curve.

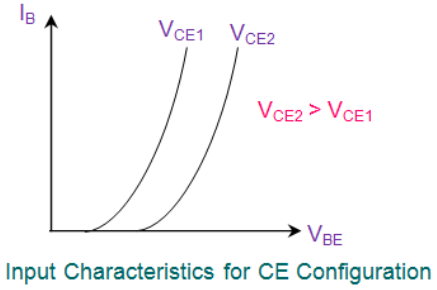


Figure 2

II. Output characteristics: It shows how collector current I_C varies with changes in V_{CE} when I_B is held constant.

1. Keep the base current I_B to some convenient value (say 20 μA) and hold it constant.

2. Increase V_{CE} from zero in steps and measure collector current I_C at each step.

3. Plot I_C (in mA) vs. V_{CE} (in volts).

4. Repeat the above experiment for different values of collector current I_B and obtain a family of curves as shown in Figure 3.

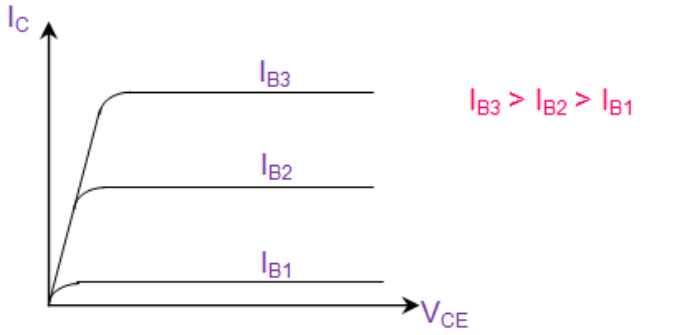


Figure 3

It is seen from the plot that as V_{CE} increases from zero, I_C rapidly increases to a saturation value. (It is observed that a small collector current flows even when base current I_B is zero, which is called common emitter leakage current, I_{CEO} which equals to $(1+\beta)I_{CO}$, where I_{CO} is due to the flow of minority carriers across the reverse biased collector base junction.)

The output impendence is defined as the ratio of variation in the collector-emitter voltage to the corresponding variation in the collector current at a constant base current:

$$R_{out} = \frac{\Delta V_{CE}}{\Delta I_C}\Big|_{I_B=\text{constant}}.$$

The value of output resistance over the nearly horizontal part of the characteristic curves varies from 10 kΩ to 50 kΩ.

III. Current transfer characteristics: It shows how collector current I_C varies with changes in I_B when V_{CE} is kept constant.

1. Keep collector emitter voltage V_{CE} constant.

2. Increase the base current I_B and measure collector current I_C.

3. Plot the base current I_B vs collector current I_C to obtain a curve as shown in Figure 4.

4. Repeat the above experiment with different values of V_{CE}.

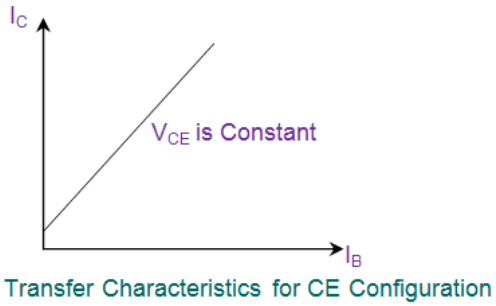


Figure 4

The current gain is defined as the ratio of a small change in the collector current to the corresponding change in the base current at constant V_{CE}:

$$\beta = \frac{\Delta I_C}{\Delta I_B}\Big|_{V_{CE}=\text{constant}} .$$

Table 1: Input Characteristics.

Sl.No.	V_{CE} = (V)		V_{CE} = (V)		V_{CE} = (V)	
	V_{BE} (V)	I_B (µA)	V_{BE} (V)	I_B (µA)	V_{BE} (V)	I_B (µA)

Table 2: Output Characteristics.

Sl.No.	I_B = (µA)		I_B = (µA)		I_B = (µA)	
	V_{CE} (V)	I_C (mA)	V_{CE} (V)	I_C (mA)	V_{CE} (V)	I_C (mA)

Table 3: Current Transfer Characteristics.

Sl.No.	V_{CE}= (V)		V_{CE}= (V)		V_{CE}= (V)	
	I_B (μA)	I_C (mA)	I_B (μA)	I_C (mA)	I_B (μA)	I_C (mA)

Finally, write a summary and comment on your results.

9. Magnetic field along the axis of the coil

Aim: Plot the graph showing the variation of magnetic field with distance along the axis of a circular coil carrying current and estimate the horizontal component of the Earth's magnetic field.

Apparatus required: Tangent galvanometer placed on a bench, DC power supply, a rheostat, a commutator, an ammeter, plug key and connecting wires.

Theory: The magnitude of the magnetic field B along the axis of a coil is given by

$$B = \frac{\mu_0 nir^2}{2(r^2 + x^2)^{3/2}},$$

where

n = number of turns in the coil,

r = radius of the coil,

i = current flowing in the coil,

x = distance of the point from the center of the coil and

μ_0 = magnetic permeability in vacuum.

If the coil's magnetic field $\boldsymbol{B}$ is made perpendicular to $\boldsymbol{B}_E$, the horizontal component of the earth's magnetic field, the deflection θ of the needle is given by, $B = B_E \tan\theta$. Thus, one has

$$B = \frac{\mu_0 nir^2}{2(r^2 + x^2)^{3/2}} = B_E \tan\theta.$$

Description of the apparatus:

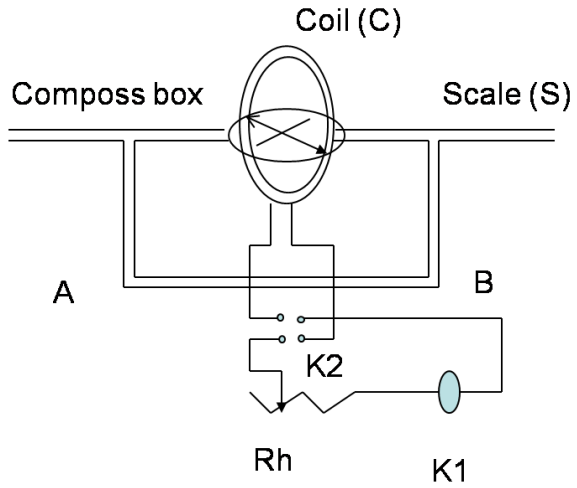


Figure 1: Schematic of the experiment

The experimental setup is shown in Figure 1. The magnetic field source is a circular coil of many turns of insulated thin copper wire. It is fixed with its plane vertical on a horizontal bench. A magnetometer compass box is placed inside the coil such that it can slide on the bench in such a way that the center of the compass always lies on the axis of the coil. The distance of the needle from the center can be read on the graduated scale fixed on the arms of the magnetometer.

Procedure:

1. Place the magnetometer box on the sliding bench such that its magnetic needle is at the centre of the coil. By rotating the whole apparatus in the horizontal plane, set the coil in the magnetic meridian. In this case the coil, needle and its image all lie in the same vertical plane. Rotate the compass box until the pointer ends read 0-0 on the circular scale.
2. To set the coils exactly in the same magnetic meridian, set up the electrical connections as in the figure. Pass the current in one direction by the help of commutator. Note the readings of the ends of the pointer. Reverse the current direction. Note the readings again. If both the readings are the same, then the coil is in the magnetic meridian. Otherwise adjust again.
3. Assign sensible least counts to all readings as well as constants such as radius of the coil, the current, the displacement and deflections.
4. Using the rheostat, adjust the current such that a deflection of 50° to 70° is produced in the compass needle placed at the centre of the coil. Read both ends of the pointer. Reverse the direction of the current and again note the readings. The mean of four readings gives the mean deflection at $x = 0$.
5. Now displace the compass box through $1 - 4$ cm such that the change in deflection is around $4 - 5°$ and repeat the experiment keeping the current unchanged throughout. Carry out the experiment for displacements on both sides of the coil assigning negative displacement to one side.
6. Plot a graph taking displacement along the x-axis and $\tan\theta$ along the y-axis with the centre of the coil as the origin. Note that this graph reveals the variation in magnetic field on the axis of the coil, up to a constant scale factor related to B_E.
7. Next compute the magnetic field B for each deflection/displacement using the parameters of the current-carrying coil.
8. Estimate the horizontal component of the Earth's magnetic field from the tabulated values of B and $\tan\theta$. Estimate the errors in F as well as $\tan\theta$ and determine the compounded uncertainty in B_E.
9. Plot the magnetic field B vs $\tan\theta$ (on the x-axis) and visually compare the data against a straight line with slope taken to be your measured value of B_E.

Table 1

x	Deflection on east arm (degrees)						Deflection on west arm (degrees)					
	Current in one way		Reverse		Mean	$\tan\theta$	Current in one way		Reverse		Mean	tanθ

where θ is the mean of four deflections for the east and west arms respectively.

Table 2

$\tan\theta$	$\delta\tan\theta$	B	δB	B_E	δB_E